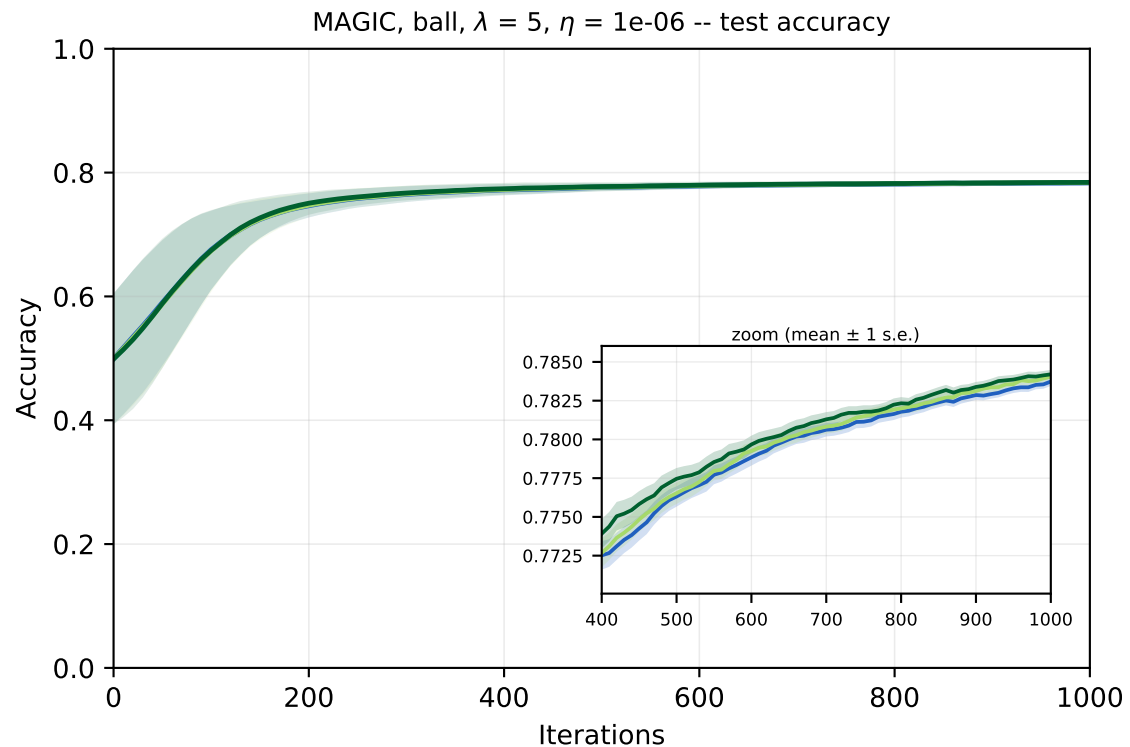
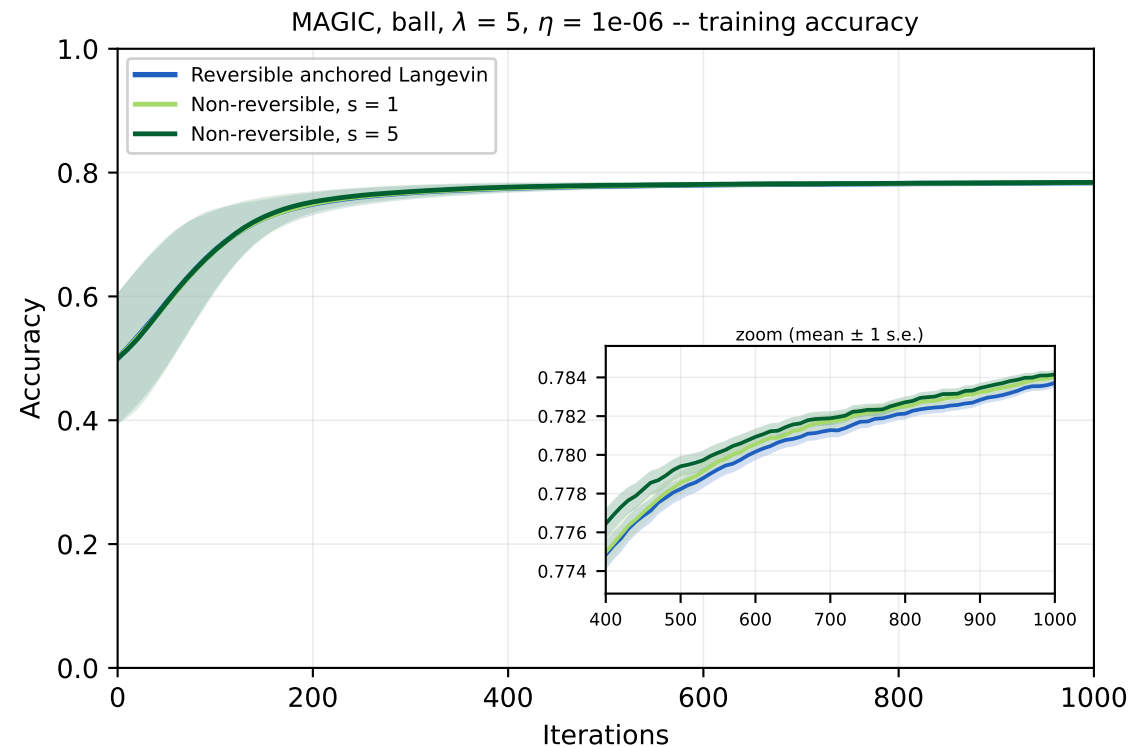


magic_ball: $\lambda = 5$, $\eta = 1\text{e-}06$



MAGIC -- ball $\|w\|^2 \leq 2$ (radius $\sqrt{2}$). $d = 10$ (intercept w_0 + 9 features). $U = \sum \text{softplus}(-y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j \geq 1} |w_j|$; U_0 replaces $|w_j|$ by $\sqrt{w_j^2 + \delta^2}$; $a = \exp(U - U_0)$. EXACT gradient. $\sigma = 10$. $\lambda_{\text{lasso}} = 5$ with $\delta = \log_2 / (9 \lambda) = 0.0154$, so a stays in $[0.50, 1]$ at every λ (raising λ also sharpens the smoothing -- the two are coupled by that choice). Step $\eta = 1\text{e-}06$. $R = 100$ independent replicates on one fixed stratified 80/20 split ($n_{\text{train}} = 15216$, $n_{\text{test}} = 3804$). Bands: mean ± 1 SD ($\text{ddof} = 1$) -- repeat-run variability at a fixed split, not a confidence or credible interval. All λ are reported in `results/lambda/lambda_lasso.csv`.